Multivariate Data Analysis - MESIO UPC

Review of previous concepts

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Presentation outline

- Vectors, scalar product, linear dependence, and projection.
- Matrices operations, special type of matrices, determinant, inverse matrix, rank, and trace.
- Eigenvalues and eigenvectors. The spectral decomposition.
- Quadratic forms, Sylvester criterion, and distances.
- Some matrix calculus.
- Random vectors and matrices.

Vectors

Vector Notation:

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \quad \mathbf{x}^\top = \begin{pmatrix} x_1, x_2, \ldots, x_n \end{pmatrix}$$

Scalar Multiplication:

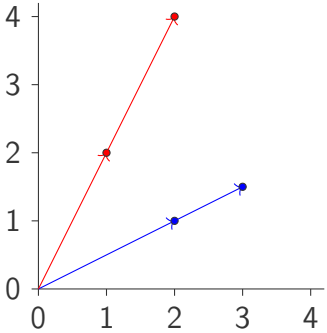
$$\alpha\mathbf{x} = \begin{pmatrix} \alpha x_1 \\ \alpha x_2 \\ \vdots \\ \alpha x_n \end{pmatrix}$$

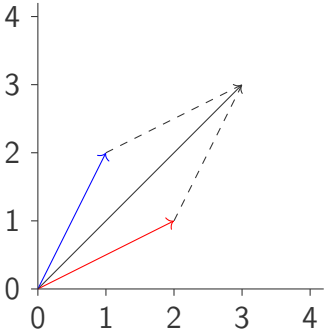
Vectors

Addition:

$$\mathbf{x} + \mathbf{y} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix} + \begin{pmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{pmatrix} = \begin{pmatrix} x_1 + y_1 \\ x_2 + y_2 \\ \vdots \\ x_n + y_n \end{pmatrix}$$

Geometric interpretation





Vectors

- **Norm**

$$\|\mathbf{x}\| = \sqrt{\mathbf{x}^\top \mathbf{x}} = \sqrt{x_1^2 + x_2^2 + \cdots + x_n^2}$$

$$\|\alpha \mathbf{x}\| = \alpha \|\mathbf{x}\|$$

- **Scalar product or Inner product**

$$\langle \mathbf{x}, \mathbf{y} \rangle = x_1 y_1 + x_2 y_2 + \cdots + x_n y_n$$

$$\langle \mathbf{x}, \mathbf{y} \rangle = 0 \Leftrightarrow \mathbf{x} \text{ and } \mathbf{y} \text{ orthogonal}$$

- **Angle**

$$\cos\theta = \frac{\langle \mathbf{x}, \mathbf{y} \rangle}{\|\mathbf{x}\| \|\mathbf{y}\|}$$

Linear combination and independence

Linear combination of n vectors:

$$\mathbf{y} = \alpha_1 \mathbf{x}_1 + \alpha_2 \mathbf{x}_2 + \cdots + \alpha_n \mathbf{x}_n$$

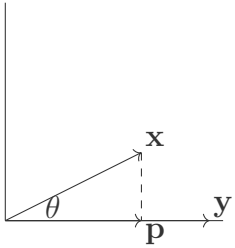
Having:

$$\alpha_1 \mathbf{x}_1 + \alpha_2 \mathbf{x}_2 + \cdots + \alpha_n \mathbf{x}_n = \mathbf{0} \tag{1}$$

Then:

- The set of vectors $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n$ is **linearly dependent** iff Eq. (1) holds for some set $(\alpha_1, \alpha_2, \ldots, \alpha_n)$ not all zero. Non-trivial representation of $\mathbf{0}$
- The set of vectors $\mathbf{x}_1, \mathbf{x}_2, \ldots, \mathbf{x}_n$ is **linearly independent** iff Eq. (1) holds only for $(\alpha_1, \alpha_2, \ldots, \alpha_n) = (0, 0, \ldots, 0)$. This is the **trivial representation** of $\mathbf{0}$

Projection



$$\cos\theta = \frac{\|\mathbf{p}\|}{\|\mathbf{x}\|}, \quad \|p\| = \frac{\langle \mathbf{x}, \mathbf{y}\rangle}{\|\mathbf{y}\|}, \quad \mathbf{p} = \alpha\mathbf{y}, \quad \alpha = \frac{\|\mathbf{p}\|}{\|\mathbf{y}\|} \rightarrow \mathbf{p} = \frac{\langle \mathbf{x}, \mathbf{y}\rangle}{\langle \mathbf{y}, \mathbf{y}\rangle}\mathbf{y}$$

Matrix

$$\mathbf{X}_{n\times p} = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{pmatrix}$$

Basic operations

- Sum

$$\mathbf{C}_{n\times p} = \mathbf{A}_{n\times p} + \mathbf{B}_{n\times p} \qquad c_{ij} = a_{ij} + b_{ij}$$

- Scalar Multiplication

$$\mathbf{C} = \alpha \mathbf{A}_{n\times p} \qquad c_{ij} = \alpha a_{ij}$$

- Product

$$\mathbf{C}_{n\times p} = \mathbf{A}_{n\times k}\mathbf{B}_{k\times p} \qquad c_{ij} = \sum_{l=1}^{k} a_{il} b_{lj}$$

Basic operations

- Transposition

$$\mathbf{C} = \mathbf{A}^\top \qquad c_{ij} = a_{ji}$$

- Transpose of the product

$$\mathbf{E} = \mathbf{AB} \qquad \mathbf{E}^\top = (\mathbf{AB})^\top = \mathbf{B}^\top \mathbf{A}^\top$$

- Inversion

$$\mathbf{A}_{k\times k} \qquad \mathbf{AB} = \mathbf{BA} = \mathbf{I} \qquad \mathbf{B}_{k\times k} = \mathbf{A}^{-1}$$

Matrix multiplication special cases

- $\mathbf{A}_{p\times k}\mathbf{B}_{k\times l} = \mathbf{C}_{p\times l}$
- $\mathbf{A}_{p\times k}\mathbf{x}_{k\times 1} = \mathbf{y}_{p\times 1}$
- $\mathbf{x}_{1\times p}^{\top}\mathbf{A}_{p\times k} = \mathbf{y}_{1\times k}$
- Given $\mathbf{D}_{n\times n}$ a diagonal matrix, then:
  - $\mathbf{X}_{n\times p}\mathbf{D}_{p\times p} = \begin{pmatrix} \mathbf{x}_1 d_1 & | & \mathbf{x}_2 d_2 & | & \cdots & | & \mathbf{x}_p d_p \end{pmatrix}$
  - $\mathbf{D}_{n\times n}\mathbf{X}_{n\times p} = \begin{pmatrix} d_1\mathbf{x}_1^{\top} \\ \hline d_2\mathbf{x}_2^{\top} \\ \hline \vdots \\ \hline d_n\mathbf{x}_n^{\top} \end{pmatrix}$

Special matrices

$$\mathbf{I_n} = \begin{pmatrix} 1 & 0 & \cdots & 0 \\ 0 & 1 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1 \end{pmatrix}, \quad \mathbf{0_n} = \begin{pmatrix} 0 & 0 & \cdots & 0 \\ 0 & 0 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 0 \end{pmatrix}$$

$$\mathbf{1_n} = \begin{pmatrix} 1 & 1 & \cdots & 1 \\ 1 & 1 & \cdots & 1 \\ \vdots & \vdots & \ddots & \vdots \\ 1 & 1 & \cdots & 1 \end{pmatrix}, \quad \mathbf{D_n} = \begin{pmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{pmatrix}$$

Symmetric and Orthogonal matrices

- Symmetric matrix: $\mathbf{A} = \mathbf{A}^\top$
- Symbolic representation:

$$\begin{pmatrix} a & b & c \\ b & d & e \\ c & e & f \end{pmatrix}$$

- Orthogonal (orthonormal) matrix:

$$\mathbf{A}\mathbf{A}^\top = \mathbf{A}^\top\mathbf{A} = \mathbf{I}_n \Leftrightarrow \mathbf{A}^\top = \mathbf{A}^{-1}$$

Determinant

Matrix property:

$$|\mathbf{A}_{k\times k}| = \sum_{j=1}^{k} a_{ij}|\mathbf{A}_{ij}|(-1)^{i+j} \qquad \text{Laplace expansion}$$

$$\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix} \qquad |\mathbf{A}| = ad - bc$$

If $k > 2$, then use a computer.

Note:

$|\mathbf{A}| = 0$ implies linear dependence, and $\mathbf{A}$ singular.

Determinant

Compute the determinant of

$$\mathbf{A} = \begin{pmatrix} 3 & 2 & 6 \\ 4 & 1 & 0 \\ 5 & 4 & 1 \end{pmatrix}$$

Command in R:

```
X <- matrix(c(3,2,6,4,1,0,5,4,1),ncol=3, byrow=TRUE)
det(X)
```

Inverse matrix computation

$$\mathbf{A}_{k\times k} \quad \mathbf{AB} = \mathbf{BA} = \mathbf{I} \quad \mathbf{B}_{k\times k} = \mathbf{A}^{-1}$$

Case 2×2:

$$\mathbf{A} = \begin{pmatrix} a & b \\ c & d \end{pmatrix}$$

$$\mathbf{A}^{-1} = \frac{1}{|\mathbf{A}|} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix} = \frac{1}{ad - bc} \begin{pmatrix} d & -b \\ -c & a \end{pmatrix}$$

If $k > 2$, then use a computer.

Inverse matrix of diagonal matrices

Inverse of a diagonal matrix:

$$\mathbf{D} = \begin{pmatrix} d_1 & 0 & \cdots & 0 \\ 0 & d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & d_n \end{pmatrix}$$

$$\mathbf{D}^{-1} = \begin{pmatrix} 1/d_1 & 0 & \cdots & 0 \\ 0 & 1/d_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & 1/d_n \end{pmatrix}$$

Matrix Rank

- **Row rank** is the maximum number of linearly independent rows of a matrix.
- **Column rank** is the maximum number of linearly independent columns of a matrix.
- The **matrix rank** is $k = \min\{\text{row rank},\ \text{column rank}\}$
- A matrix of rank k can be represented exactly in a k dimensional space.
- A **full-rank matrix** is a square matrix in which the **column rank** and **row rank** both equal to the **dimension** n of the matrix.

Trace

$$\mathbf{A}_{k\times k} \qquad tr(\mathbf{A}) = \sum_{i=1}^{k} a_{ii}$$

Some properties:

- $tr(\alpha\mathbf{A}) = \alpha tr(\mathbf{A})$
- $tr(\mathbf{AB}) = tr(\mathbf{BA}) \qquad tr(\mathbf{ABC}) = tr(\mathbf{CAB}) = tr(\mathbf{BCA})$
- $tr(\mathbf{AA}^\top) = \sum_{i=1}^{k}\sum_{j=1}^{k} a_{ij}^2$ (Sum-of-squares, a measure of variability in a covariance matrix)

Eigenvalues and eigenvectors

$$\mathbf{A}\mathbf{v} = \lambda\mathbf{v} \qquad \mathbf{A}_{k\times k}$$

$$(\mathbf{A} - \lambda\mathbf{I}_n)v = 0$$

The characteristic equation is

$$|\mathbf{A} - \lambda\mathbf{I}_n| = 0$$

- There are k roots, not necessarily all distinct.
- If $\mathbf{A} = \mathbf{A}^\top$, all roots are real.
- Each root (eigenvalue) has an associated eigenvector.
- $\mathbf{v}$ is usually scaled (normalized) to unit length such that $\mathbf{v}^\top\mathbf{v} = 1$.

Spectral decomposition

if $\mathbf{A}_{k\times k}$ and $\mathbf{A} = \mathbf{A}^\top$ and $\|\mathbf{v_i}\| = 1$ then

$$\mathbf{A} = \sum_{i=1}^{k} \lambda_i \mathbf{v}_i \mathbf{v}_i^\top = \lambda_1 \mathbf{v}_1 \mathbf{v}_1^\top + \lambda_2 \mathbf{v}_2 \mathbf{v}_2^\top + \cdots + \lambda_k \mathbf{v}_k \mathbf{v}_k^\top$$

$$\mathbf{A} = \mathbf{V}\mathbf{D}_\lambda \mathbf{V}^\top$$

$$\mathbf{V} = (\mathbf{v}_1|\mathbf{v}_2|\cdots|\mathbf{v}_k), \quad \mathbf{D}_\lambda = \begin{pmatrix} \lambda_1 & 0 & \cdots & 0 \\ 0 & \lambda_2 & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & \lambda_k \end{pmatrix}, \quad \mathbf{V}^\top \mathbf{V} = \mathbf{I}.$$

Spectral decomposition

- Eigenvalues are usually ordered such that $\lambda_1 \geq \lambda_2 \geq \lambda_3 \geq \cdots \geq \lambda_k$.
- If $\mathbf{A}$ is not of full rank, it will have at least one eigenvalue equal to 0.
- $tr(\mathbf{A}) = tr(\mathbf{V}\mathbf{D}_\lambda\mathbf{V}^\top) = \mathrm{tr}(\mathbf{V}^\top\mathbf{V}\mathbf{D}_\lambda) = tr(\mathbf{D}_\lambda) = \sum_{i=1}^k \lambda_i$.
- $|\mathbf{A}| = |\mathbf{V}\mathbf{D}_\lambda\mathbf{V}^\top| = |\mathbf{V}||\mathbf{D}|_\lambda|\mathbf{V}|^t = |\mathbf{V}||\mathbf{V}|^t|\mathbf{D}|_\lambda = |\mathbf{V}\mathbf{V}^\top\mathbf{D}_\lambda| = |\mathbf{D}_\lambda| = \prod_{i=1}^k \lambda_i$
- $\lambda_1\mathbf{v}_1\mathbf{v}_1^\top + \lambda_2\mathbf{v}_2\mathbf{v}_2^\top$ provides a rank 2 least squares approximation to $\mathbf{A}$ (the best one).

Exercise

Given

$$\mathbf{S} = \begin{pmatrix} 1 & 2 \\ 2 & 1 \end{pmatrix}$$

Compute the eigenvalues and eigenvectors by hand and check the result with R. Is S a covariance matrix?

Command in R:

```
X <- matrix(c(1,2,2,1),ncol=2, byrow=TRUE)
eigen(X)
```

Matrix powers

$$\mathbf{A} = \mathbf{V}\mathbf{D}\mathbf{V}^\top$$

$$\mathbf{A}^2 = \mathbf{V}\mathbf{D}\mathbf{V}^\top\mathbf{V}\mathbf{D}\mathbf{V}^\top = \mathbf{V}\mathbf{D}^2\mathbf{V}^\top$$

$$\mathbf{A}^k = \mathbf{V}\mathbf{D}^k\mathbf{V}^\top$$

$$\mathbf{A}^{-1} = \mathbf{V}\mathbf{D}^{-1}\mathbf{V}^\top$$

$$\mathbf{A}^{\frac{1}{2}} = \mathbf{V}\mathbf{D}^{\frac{1}{2}}\mathbf{V}^\top$$

$$\mathbf{A}^{-\frac{1}{2}} = \mathbf{V}\mathbf{D}^{-\frac{1}{2}}\mathbf{V}^\top$$

Quadratic forms

$$4x_1^2 + 5x_2^2 + 3x_3^2 + 2x_1x_2 + 4x_1x_3 + x_2x_3 = \mathbf{x}^\top \mathbf{A}\mathbf{x}$$

$$\mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ x_3 \end{pmatrix} \qquad \mathbf{A} = \begin{pmatrix} 4 & 1 & 2 \\ 1 & 5 & \frac{1}{2} \\ 2 & \frac{1}{2} & 3 \end{pmatrix}$$

Quadratic forms

In general:

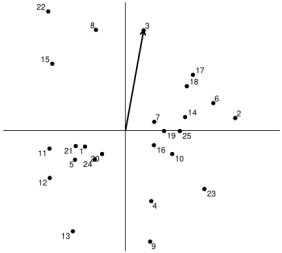
$$\sum_{i=1}^{k}\sum_{j=1}^{k} a_{ij}x_i x_j = \mathbf{x}^\top \mathbf{A}\mathbf{x}$$

- $\mathbf{A}$ is **positive definite** if $\mathbf{x}^\top \mathbf{A}\mathbf{x} > 0$ for all $\mathbf{x} \neq \mathbf{0}$.
- $\mathbf{A}$ is **positive semi-definite** if $\mathbf{x}^\top \mathbf{A}\mathbf{x} \geq 0$ for all $\mathbf{x} \neq \mathbf{0}$.
- $\mathbf{A}$ is **negative definite** if $\mathbf{x}^\top \mathbf{A}\mathbf{x} < 0$ for all $\mathbf{x} \neq \mathbf{0}$.
- $\mathbf{A}$ is **negative semi-definite** if $\mathbf{x}^\top \mathbf{A}\mathbf{x} \leq 0$ for all $\mathbf{x} \neq \mathbf{0}$.
- $\mathbf{A}$ is **indefinite** if it does not satisfy any of the above conditions.

Quadratic forms and eigenvalues

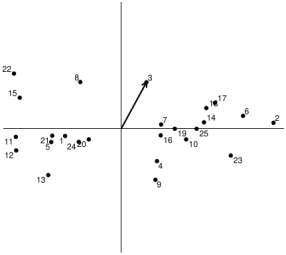
- $\mathbf{A}$ positive definite $\Leftrightarrow$ all $\lambda_i > 0$
- $\mathbf{A}$ positive semi-definite $\Leftrightarrow$ all $\lambda_i \geq 0$
- $\mathbf{A}$ negative definite $\Leftrightarrow$ all $\lambda_i < 0$
- $\mathbf{A}$ negative semi-definite $\Leftrightarrow$ all $\lambda_i \leq 0$
- $\mathbf{A}$ indefinite in all other cases.

Quadratic forms and distance



- Let $\mathbf{x}$ be a single observation written as a column vector.
- $\|\mathbf{x}\|^2 = \langle \mathbf{x}^\top, \mathbf{x} \rangle = x_1^2 + x_2^2 = \mathbf{x}^\top \mathbf{A} \mathbf{x}$ with $\mathbf{A} = \mathbf{I}_n$
- $\mathbf{x}^\top \mathbf{A} \mathbf{x}$ is the squared Euclidean distance from the origin.
- $(\mathbf{x} - \mathbf{y})^\top \mathbf{A} (\mathbf{x} - \mathbf{y})$ is the squared Euclidean distance from $\mathbf{x}$ to $\mathbf{y}$.

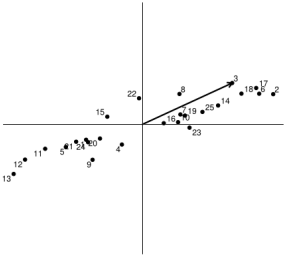
Quadratic forms and distance



Account for difference in variability: $\tilde{\mathbf{x}} = \mathbf{D}^{-1}\mathbf{x}$ $\quad \mathbf{D} = \begin{pmatrix} s_1 & 0 \\ 0 & s_2 \end{pmatrix}$

- $\|\tilde{\mathbf{x}}\|^2 = \langle \tilde{\mathbf{x}}^\top, \tilde{\mathbf{x}} \rangle = \left(\frac{x_1}{s_1}^2\right) + \left(\frac{x_2}{s_2}^2\right) = \mathbf{x}^\top \mathbf{A} \mathbf{x}$ with $\mathbf{A} = \mathbf{D}^{-2} = \begin{pmatrix} \frac{1}{s_1^2} & 0 \\ 0 & \frac{1}{s_2^2} \end{pmatrix}$
- $\mathbf{x}^\top \mathbf{A} \mathbf{x}$ is squared Weighted Euclidean distance from the origin.
- $(\mathbf{x} - \mathbf{y})^\top \mathbf{A} (\mathbf{x} - \mathbf{y})$ is the squared Weighted Euclidean distance from $\mathbf{x}$ to $\mathbf{y}$.

Quadratic forms and distance



Account for difference in variability and for correlation: $\hat{\mathbf{x}} = \mathbf{S}^{-\frac{1}{2}}\mathbf{x}$. The Mahalanobis transformation.

- $\|\hat{\mathbf{x}}\|^2 = \langle \hat{\mathbf{x}}^\top, \hat{\mathbf{x}} \rangle = \mathbf{x}^\top \mathbf{S}^{-1}\mathbf{x} = \mathbf{x}^\top \mathbf{A}\mathbf{x}$ with $\mathbf{A} = \mathbf{S}^{-1} = \begin{pmatrix} s_1^2 & s_{12} \\ s_{21} & s_2^2 \end{pmatrix}^{-1}$
- $\mathbf{x}^\top \mathbf{A}\mathbf{x}$ is the squared Mahalanobis distance from the origin.
- $(\mathbf{x} - \mathbf{y})^\top \mathbf{A}(\mathbf{x} - \mathbf{y})$ is the squared Mahalanobis distance from $\mathbf{x}$ to $\mathbf{y}$.

Geometry of eigenvalues and eigenvectors

- The equation $x_1^2 + x_2^2 = c^2$ represents a circle, the equation $a_1 x_1^2 + a_2 x_2^2 = c^2$ represents an ellipse and $a_{11} x_1^2 + 2a_{12} x_1 x_2 + a_2 x_2^2 = c^2$ represents an ellipse with a rotation.

- A quadratic form gives squared distances from origin: $\mathbf{x}^\top \mathbf{A} \mathbf{x} = c^2$

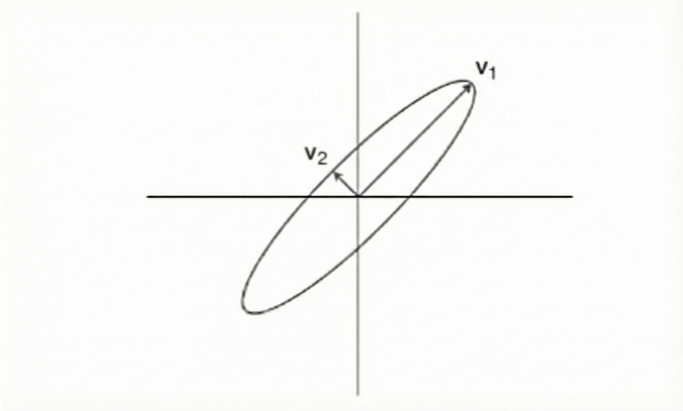
- Expanding $\mathbf{A}$ using its spectral decomposition we get:

$$\mathbf{x}^\top \mathbf{A} \mathbf{x}^\top = \mathbf{x} \mathbf{V} \mathbf{D}_\lambda \mathbf{V}^\top \mathbf{x} = \mathbf{x}^\top (\lambda_1 \mathbf{v}_1 \mathbf{v}_1^\top + \lambda_2 \mathbf{v}_2 \mathbf{v}_2^\top) \mathbf{x} =$$
$$\lambda_1 (\mathbf{x}^\top \mathbf{v}_1)^2 + \lambda_2 (\mathbf{x}^\top \mathbf{v}_2)^2 = \lambda_1 \alpha^2 + \lambda_2 \alpha^2 = c^2$$

- $\mathbf{x}^\top \mathbf{v}_i$ is the projection of vector $\mathbf{x}$ on vector $\mathbf{v}_i$

- This means that $\mathbf{x}$ is transformed by $\mathbf{A}$ in such a way that its components are projected along the eigenvectors directions and scaled inversely by: $\sqrt{\frac{c}{\lambda_i}}$

Geometry of eigenvalues and eigenvectors



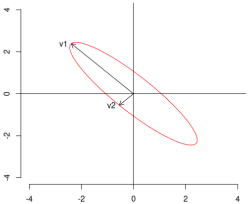
Ellipse geometry induced by eigenvectors in a quadratic form

Example

$$\mathbf{S} = \begin{pmatrix} 1.00 & -0.90 \\ -0.90 & 1.00 \end{pmatrix} \qquad \lambda_1 = 1.9 \quad \lambda_2 = 0.1$$

$$\mathbf{V} = \begin{pmatrix} -0.7071 & -0.7071 \\ 0.7071 & -0.7071 \end{pmatrix} \qquad \mathbf{x}^\top \mathbf{S}^{-1} \mathbf{x} = 2.4474^2 = 5.99$$

Half-length of principal axis: $2.4474 \cdot \sqrt{1.9} = 3.374$; $\quad 2.4474 \cdot \sqrt{0.1} = 0.774$



More on the Mahalanobis distance

- The Mahalanobis distance follows a χ^2_p distribution with p degrees of freedom.
- For $\alpha = 0.05$ we have an ellipse that covers the 95% of the observations. This is a generalization of **confidence interval**.

$$P[((\mathbf{x}-\mu)^\top \mathbf{S}^{-1}(\mathbf{x}-\mu) \leq \chi^2_2] = 0.95 \Rightarrow c^2 = 5.99$$

- Calculation of the Mahalanobis distance with the original observations is equal to the Euclidean distance with standardized observations if variables are uncorrelated.
- **Mahalanobis transformation**: $z = S^{-1/2}(x-\mu)$.
- The Mahalanobis transformation **decorrelates the data**.

Matrix calculus

$$f(\mathbf{x}) = \mathbf{a}^\top \mathbf{x} = a_1 x_1 + a_2 x_2 + \cdots + a_p x_p$$

$$\frac{\partial f}{\partial x_1} = a_1, \quad \frac{\partial f}{\partial x_2} = a_2, \cdots, \quad \frac{\partial f}{\partial x_p} = a_p$$

$$\frac{\partial f(\mathbf{x})}{\partial \mathbf{x}} = \mathbf{a} \quad \text{the gradient vector}$$

$$f(\mathbf{x}) = \mathbf{A}\mathbf{x} \quad \frac{\partial f(\mathbf{x})}{\partial \mathbf{x}} = \mathbf{A}^\top$$

$$f(\mathbf{x}) = \mathbf{x}^\top \mathbf{A}\mathbf{x} \quad \frac{\partial f(\mathbf{x})}{\partial \mathbf{x}} = (\mathbf{A} + \mathbf{A}^\top)\mathbf{x}$$

$$\text{If } (\mathbf{A} = \mathbf{A}^\top) \Rightarrow \frac{\partial f(\mathbf{x})}{\partial \mathbf{x}} = 2\mathbf{A}\mathbf{x}$$

Random vectors and matrices

$$\mathbf{X} = \begin{pmatrix} X_1 \\ X_2 \\ X_3 \\ \vdots \\ X_p \end{pmatrix} \qquad \mathbb{E}[\mathbf{X}_{p\times 1}] = \begin{pmatrix} \mathbb{E}[X_1] \\ \mathbb{E}[X_2] \\ \mathbb{E}[X_3] \\ \vdots \\ \mathbb{E}[X_p] \end{pmatrix}$$

$$\mathbb{E}[\mathbf{A}\mathbf{X}] = A\mathbb{E}[\mathbf{X}]$$

$$\mathbb{E}[\mathbf{X} + \mathbf{Y}] = \mathbb{E}[\mathbf{X}] + \mathbb{E}[\mathbf{Y}]$$

Population mean vector

$$\mathbb{E}[\mathbf{X}] = \boldsymbol{\mu} = \begin{pmatrix} \mu_1 \\ \mu_2 \\ \vdots \\ \mu_p \end{pmatrix}$$

Random vectors and matrices

Population covariance matrix

$$Cov(\mathbf{X}) = \mathbf{\Sigma}_{p\times p} = \mathbb{E}[(\mathbf{X}-\boldsymbol{\mu})(\mathbf{X}-\boldsymbol{\mu})^\top] = \begin{pmatrix} \sigma_{11} & \sigma_{12} & \cdots & \sigma_{1p} \\ \sigma_{12} & \sigma_{22} & \cdots & \sigma_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{1p} & \sigma_{2p} & \cdots & \sigma_{pp} \end{pmatrix}$$

$$\mathbf{Y} = \mathbf{A}\mathbf{X}$$

$$Cov(\mathbf{Y}) = Cov(\mathbf{A}\mathbf{X}) = \mathbf{A}\mathbf{\Sigma}_{\mathbf{p}\times\mathbf{p}}\mathbf{A}^\top$$

Sample data matrices

- Data matrix $\mathbf{X}_{n\times p}$
- Sample mean vector $\mathbf{m}_{p\times 1} = (\frac{1}{n}\mathbf{1}^\top\mathbf{X})^\top$
- Centered data matrix

$$\mathbf{X}_c = \mathbf{X} - \mathbf{1}_{n\times 1}\mathbf{m}^\top = \mathbf{X} - \frac{1}{n}\mathbf{1}\mathbf{1}^\top\mathbf{X} = (\mathbf{I} - \frac{1}{n}\mathbf{1}\mathbf{1}^\top)\mathbf{X}$$

- Centring matrix $\mathbf{H} = \mathbf{I} - \frac{1}{n}\mathbf{1}\mathbf{1}^\top \qquad \mathbf{X}_c = \mathbf{H}\mathbf{X}$
- Standardized data matrix

$$\mathbf{X}_s = \mathbf{X}_c\mathbf{D}_s^{-1} \qquad \mathbf{D}_s = diag(s_1, s_2, ..., s_p) \qquad \mathbf{X}_s = \mathbf{H}\mathbf{X}\mathbf{D}_s^{-1}$$

- Sample covariance matrix $\mathbf{S} = \frac{1}{n-1}\mathbf{X}_c^\top\mathbf{X}_c$
- Sample correlation matrix $\mathbf{R} = \mathbf{D}_s^{-1}\mathbf{S}\mathbf{D}_s^{-1} = \frac{1}{n-1}\mathbf{X}_s^\top\mathbf{X}_s$

In-depth sample covariance matrix

$$\mathbf{S}_{n-1} = \begin{pmatrix} s_{11} & s_{12} & \cdots & s_{1p} \\ s_{12} & s_{22} & \cdots & s_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ s_{1p} & s_{2p} & \cdots & s_{pp} \end{pmatrix} = \frac{1}{n-1}\mathbf{X}_c^\top \mathbf{X}_c$$

$$= \frac{1}{n-1}\sum_{i=1}^{n}(\mathbf{x}_i - \mathbf{m})(\mathbf{x}_i - \mathbf{m})^\top$$

$$\mathbf{S}_n = \frac{n-1}{n}\mathbf{S}_{n-1}$$

In-depth sample correlation matrix

$$\mathbf{R} = \begin{pmatrix} 1 & r_{12} & \cdots & r_{1p} \\ r_{12} & 1 & \cdots & r_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ r_{1p} & r_{2p} & \cdots & 1 \end{pmatrix} = \frac{1}{n-1}\mathbf{X}_s^\top \mathbf{X}_s = \mathbf{D}_s^{-1}\mathbf{S}\mathbf{D}_s^{-1}$$

Population vs. Sample

Population

(conceptual)

Random vector $\mathbf{X}_{p\times 1}$

Population mean vector

$\boldsymbol{\mu}_{p\times 1} = \mathbb{E}[\mathbf{X}]$

Population covariance matrix

$\boldsymbol{\Sigma}_{p\times p} = \mathbb{E}[(\mathbf{X}-\boldsymbol{\mu})(\mathbf{X}-\boldsymbol{\mu})^\top]$

Sample

(observed)

Data matrix $\mathbf{X}_{n\times p}$

Sample mean vector

$\mathbf{m}_{p\times 1} = \frac{1}{n}(\mathbf{1}^\top\mathbf{X})^\top$

Sample covariance matrix

$\mathbf{S}_{p\times p} = \frac{1}{n-1}\sum_{i=1}^{n}(\mathbf{x}_i-\mathbf{m})(\mathbf{x}_i-\mathbf{m})^\top$

The duality

$$\mathbf{X}_{n\times p} = \begin{pmatrix} x_{11} & x_{12} & \cdots & x_{1p} \\ x_{21} & x_{22} & \cdots & x_{2p} \\ \vdots & \vdots & \ddots & \vdots \\ x_{n1} & x_{n2} & \cdots & x_{np} \end{pmatrix} = \begin{pmatrix} \mathbf{x}_1^\top \\ \mathbf{x}_2^\top \\ \vdots \\ \mathbf{x}_n^\top \end{pmatrix} = \left(\mathbf{y}_1 | \mathbf{y}_2 | \cdots | \mathbf{y}_p\right)$$

We will work with:

- n points in a p-dimensional space.
- p vectors in a n-dimensional space.

Measures of variance

- $|\mathbf{S}|$: Generalized variance.
- $\mathrm{tr}(\mathbf{S})$: Total variance.
- $\lambda_1, \lambda_2, \cdots, \lambda_p$: Decomposed variance.

References

- Johnson and Wichern, (2002) *Applied Multivariate Statistical Analysis*, 6th edition, Prentice Hall, Chapter 2 & 3.
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